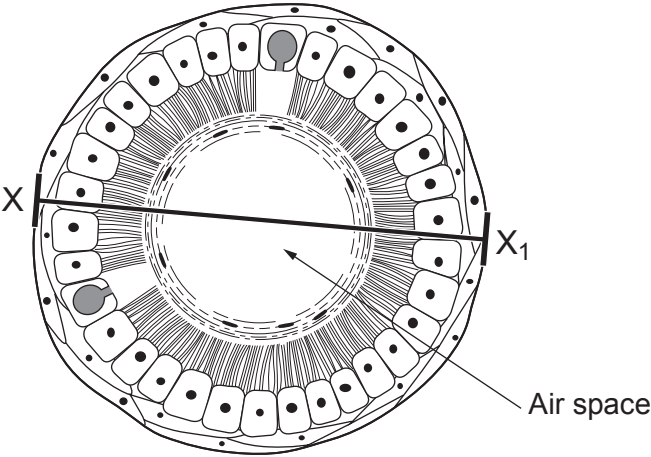


6. The diagram below shows a cross section of a healthy bronchiole.

Diagram A

A healthy bronchiole

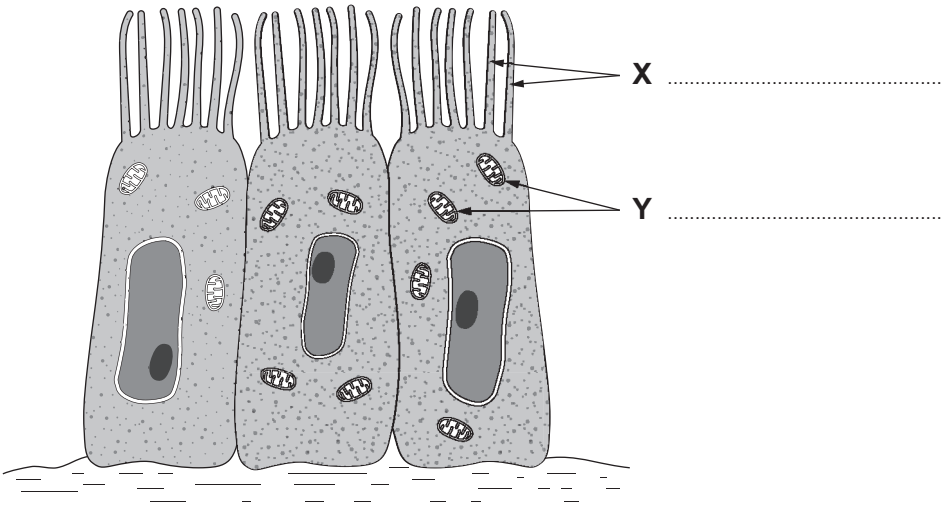


- (a) If the magnification of diagram A is x50, use the line X–X₁ to calculate the actual diameter of the bronchiole. [1]

Diameter = mm

- (b) (i) Diagram **B** below shows a highly magnified close-up of three cells of the lining of the bronchiole.
Label parts X and Y on diagram B. [2]

Diagram B



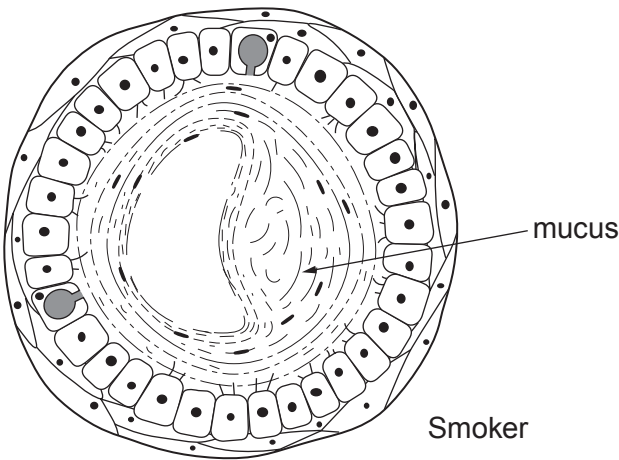
- (ii) Explain how the process taking place in part **Y** allows part **X** to carry out its function. [3]

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The following diagram shows a section through a bronchiole from a person who had smoked for 20 years.



- (c) What would you expect to find in the mucus of this person? [1]

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Examiner
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- (d) Nicotine is the addictive substance in tobacco. People can be helped to stop the habit of smoking by giving them nicotine replacement therapy (NRT). The table gives the results of the trials of different types of NRT. The placebo is a neutral substance which does not contain nicotine.

	Smokers given NRT		Smokers given a placebo	
Type of NRT	Numbers of smokers in trial	Percentage of smokers who gave up smoking	Numbers of smokers in trial	Percentage of smokers who gave up smoking
chewing gum	7708	20	9319	12
patch on skin	7387	14	5969	8
nasal spray	448	24	439	12
inhaler	490	14	486	8
tablet	243	20	245	13

- (i) Why was the placebo used in these trials? [1]

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- (ii) Use the results to state the type of NRT in which scientists would have the greatest confidence. Give a reason for your answer. [2]

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- (iii) Which type of NRT was most effective? Give a reason for your answer. [2]

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